

The Forest of Food

“Technical Specification Document”

GAM150(S20KR)

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Team name

Gathering

Team Roster

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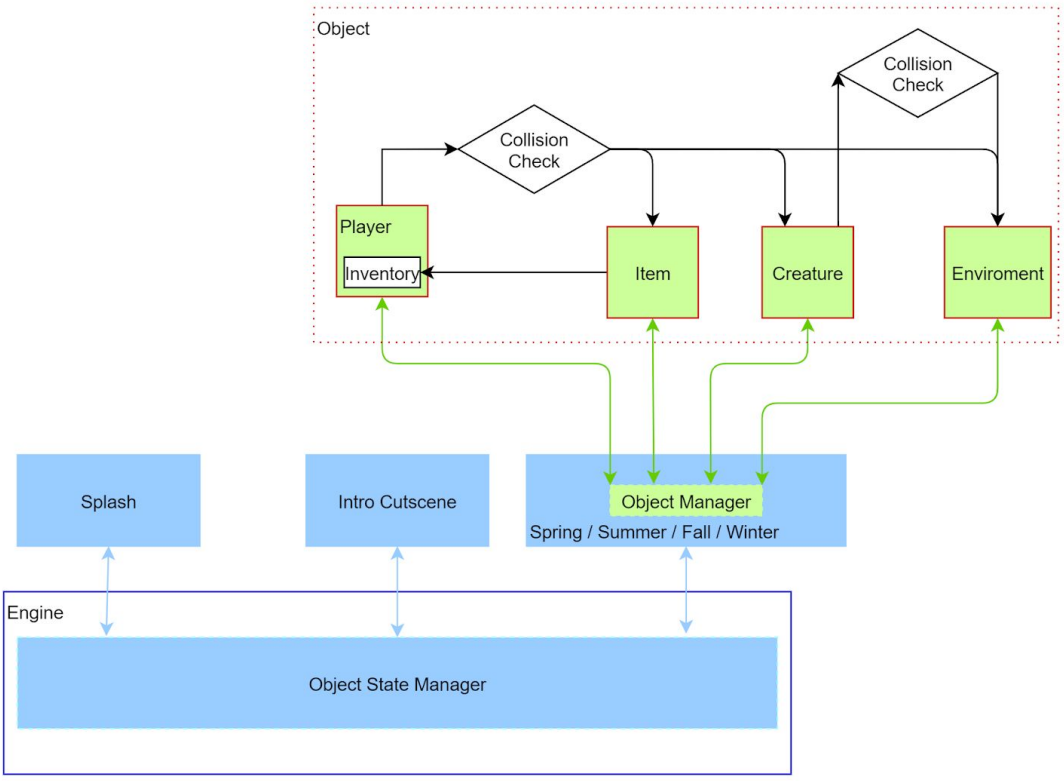
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Technical Specification Structure

Architecture Overview



Graphics Implementation

Our game follow the graphics implementation of cs230. Our game made the Matrices, Texture and sprite files in Engine filters. To use those files, we will manage the Graphics Implementation. When we start the animation, we will make the sprite sheet,spt files and anim files(cs230 style).

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Physics Implementation

Our Physics system uses acceleration method but we are planning to change velocity only movement for retro-style moving and collision detection is using square collision box. Fortunately, our game doesn't need to add gravity or bound. So we just need collision check.

Player Controls Implementation

Our game only has single player. Our game can play only depending on Keyboard.

- ***Key***

Move Left - Press Left Arrow
Move Right - Press Right Arrow
Move Up - Press Up Arrow
Move Down - Press Down Arrow
Attack - Press Space Key
Interact - Control Key
Inventory - Z key

- ***Input System***

Our game use the doodle input system also follow cs230 style.
For example, that situation will happen we might stop the playing game.



Behavior Implementation

The Forest of food don't have enemies. But there are animals and Fairy of the forest of food in game.
Animals can move freely on the map. It will be setting 2 cases.

- *case 1 : Animal can move left - right.*
- *case 2 : Animal can move up - down.*

Animals have velocity,max velocity and random position.
When the Animal spawns on the map, they have the limit movement depending on their random position. It means that managing the animal movement.

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Animal manage to use the vector. One Creature class - Animals class.

The fairy of the forest of food just will show two sections.

- *First Section : Story Section*
- *Second Section : Cooking Section , Ending Section*

Coding Methods

- *File Naming Convention*

File name should show the main function of the file.

Ex) Story_Stage

- *Filters*

Engine File should be in Engine filters.

Ex)Engine,GameObject Manager,GameObject,GameState..

Game File should be in Game filters.

Ex)Stage,Animal,Player,Map,Splash..

- *Variable Convention*

The first letter of the Variable should be small letter and
the second letter of the variable name should be the capital letter.

Ex) keyBoardButton

- *Function Convention*

The first letter or word of the function name should be the capital letter.

Ex) IsKeyPressed()

- *Class Convention*

The name of class should be the one word.

Ex) Creature

- *Bool Convention*

The name of bool type should be the question type.

Ex) isJumping

Debugging

Debugging - Use output, console and BreakPoint.
Release Mode - To shut off our Debug tools.

Tools

Visual studio (For Debugging and Coding)
Visual Studio Editor(Check the Merge error)
Aseprite (Draw Pixel Art)
Photoshop(Draw Art)

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MidiBang Paint Pro (Draw Art)
Music Maker 2020(For SFX)

Scripting Languages

Nothing.

Technical Risks

- *Risk*

Object Management Problem

We have many objects like trees,animals and items.
So we consider how to manage the all of the objects. Now the tree and animal and item class inherit Object class. But Each has a different management method.
Tree and animal class has the manager class. So I make the vector for tree object and animal object. And Item class use the linked list. Each Item save information and code number on free list. So when the game needs the that objects, get the free list.
Different Management method is very inefficient for collision check (also other things.. animation stuff..).

Too much Expensive Resource for Art files

We have a lot of art files. 4 Trees, 4 Tiles, Player sprite, 4 Animals sprites, Story Scene file. Too big resource for our game. That makes our game's memory higher.

Object's Random position

Animal Objects,Tree Objects and Item Objects spawn randomly depending on random position generator function. But Sometimes The objects overlap each other. Because they have same position on the map! So It looks like very unnatural.

- *Mitigation*

Object Management Problem

To solve this problem, we will make the object manager class because managing objects all at once. But that also makes problem. We have lots of objects. So we put together linked list in Game Object manager. When the object push back in Game Object vector, they have their unique code. That might be the better idea than now.

Too much Expensive Resource for Art files

There are two solutions for this problem.

- *First Solution: Make Asset Manager. And Use unique pointer and weak pointer on Asset Manager. On asset Manager manage all of Asset Files. Unique pointer is allocating and deallocation itself. Weak Pointer deallocate itself.*
- *Second Solution:Make Texture Manager. On texture Manager's function parameter is filename. Use std::Map function store on Map with Key. And If the texture was loaded, check on std::Map. So there are reuse the load file. So it can manage faster than before.*

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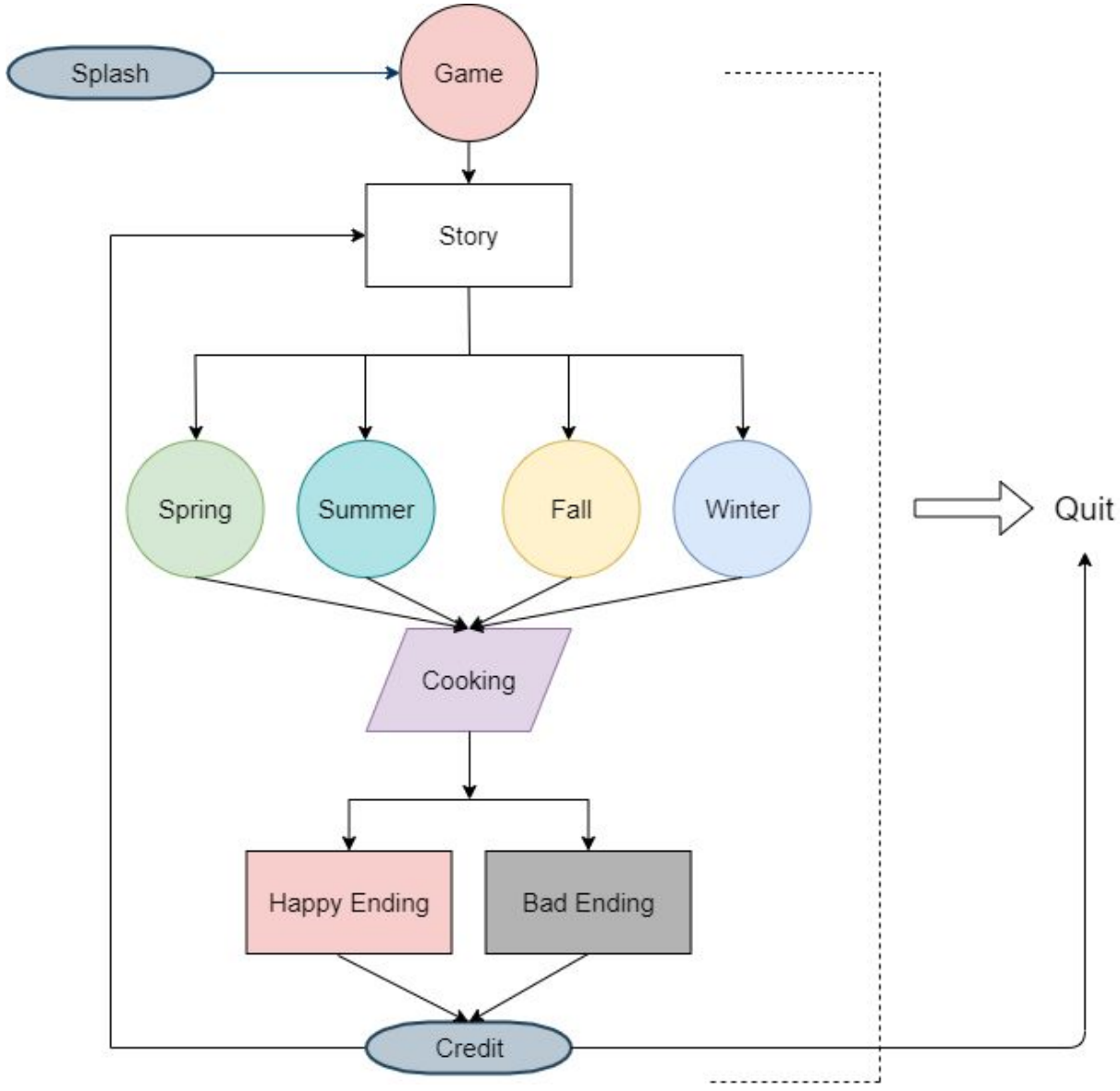
Object's Random position

Make new Random Function. Before we just use random function when they spawn. But We change Tile Base. Not doodle random used mersenne twister random. If they get Tile Number, check the existence of object around the tile. So if there are objects near tile number, make new random tile number. And Check again , again. It may help overlap each other.

Appendices

Appendix A: Interface Flow

FlowChart



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Mockups

- *Splash: Bring up the DigiPen logo.*
- *Game:*
    1. *The game will start immediately after Splash is over. In the game, the story goes first.*
    2. *The player progresses through the story and selects one season; spring, summer, fall, and winter.*
    3. *Players cook with ingredients from each season.*
    4. *Then the player can see the happy ending or the bad ending.*
    5. *Credits come out and go back to the story or the game ends.*
- *Quit: Press the ESC key at any time to end the game.*

Appendix B: Art Requirements

- *All files must be in .png format.*
- *The file name should be accurate to what that file represents. For example, if there is rice, you should use* **Rice.png.** *The first letter must be uppercase.*
- *The source of all art was done by programmer.*
- *The art used in the story is an illustration, and the art used in the game is pixel art. Ex)*



Story art



In game art

Appendix C: Audio Requirements

- *Audio has different BGM(Background Music) depending on spring, summer, fall, and winter.*
- *All audio is either programmer make or is from the DigiPen library.*
- *When a knife and animals collide, when picking up an item, and when opening inventory, the sound effect is generated.*

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